

SEBI Grade A [IT] – PYQs – 2

(Memory Based)

Q. Which of the following statements is true for Minimum Spanning Tree (MST)?

- A) MST is unique for all graphs
- B) MST may not be unique if equal edge weights exist
- C) MST always includes all edges
- D) MST always has maximum weight

```
int fact(int n){  
    if(n<=1) return 1;  
    else return n*fact(n-1);  
}  
int main(){  
    printf("%d", fact(4));  
}
```

Q. What will be the output of the above program?

- A) 12
- B) 24
- C) 16
- D) 20

Q. Which of the following is TRUE about firewalls?

- A) Firewalls prevent all types of attacks
- B) Firewalls can block unauthorized traffic based on rules
- C) Firewalls encrypt all transmitted data
- D) Firewalls replace antivirus software

Q. Which UNIX command is used to list all files including hidden files in the current directory?

- A) ls -l
- B) ls -h
- C) **ls -a**
- D) ls -al



Q. Which of the following statements about an Independent Data Mart is TRUE?

- A) It requires a data warehouse as its source
- B) It is faster to implement but may cause data inconsistency
- C) It always shares metadata with other marts
- D) It follows the top-down approach of Inmon

Q. Which command is used to display the first 10 lines of a file?

- A) tail
- B) **head**
- C) more
- D) less

Q. What will be printed by this script?

```
a=5  
if [ $a -gt 3 ]; then  
    echo "Greater"  
else  
    echo "Smaller"  
fi
```

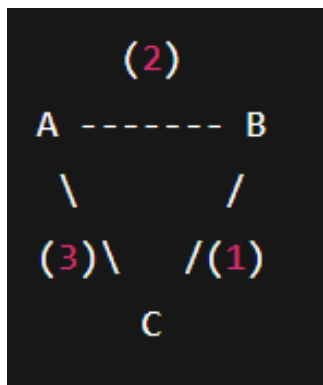
- A) Greater
- B) Smaller
- C) Error
- D) Nothing

Q. Which shell variable stores the exit status of the last executed command?

- A) **\$?**
- B) \$\$
- C) \$#
- D) \$!



Q. Find the total cost of the MST using Kruskal's Algorithm.



- A) 3
- B) 4
- C) 2
- D) 5

Q. Which of the following loops executes until a given condition becomes true?

- A) while
- B) for
- C) **until**
- D) repeat

Q. Which OLAP operation involves viewing data from different perspectives?

- A) Drill-down
- B) Roll-up
- C) Slice and Dice
- D) **Pivot**



Q. What is the exit status of a successful command in UNIX?

- A) -1
- B) 1
- C) 0
- D) 255



Q. Which type of data mart is typically created for a specific business unit before the full data warehouse is implemented?

- A) Hybrid Data Mart
- B) Dependent Data Mart
- C) **Independent Data Mart**
- D) Operational Data Store

Q. DNS uses which transport protocol by default and why?

- A) TCP only, because DNS prefers reliability
- B) UDP only, because queries are small
- C) UDP for queries; TCP for zone transfers
- D) TCP for queries; UDP for replies



Q. Which protocol retrieves emails from the server without leaving a copy by default?

- A) **POP3**
- B) IMAP
- C) SMTP
- D) NNTP



Q. A foreign key referencing a non-candidate key violates:

- A) Entity integrity
- B) Referential integrity
- C) **Key constraint**
- D) Not-null constraints

Q. Choose the relational algebra expression that gives employees who work in ALL projects of dept 10:

- A) $\pi(\text{Ename})(\text{Emp} \bowtie \text{WorksOn} \bowtie \sigma(\text{Dept}=10)(\text{Project}))$
- B) $\pi(\text{Ename})(\text{Emp} \div \sigma(\text{Dept}=10)(\text{Project}))$
- C) $\sigma(\text{Ename})(\text{Emp} \times \text{WorksOn} \times \text{Project})$
- D) $\pi(\text{Ename})(\text{Emp} \bowtie \text{Project})$

Q. Which is TRUE for a B+ Tree but NOT for a B-Tree?

- A) Keys exist only in leaf nodes
- B) Height is always 1
- C) Only one child per internal node
- D) Parent keys contain data

Q. What happens if a derived class has a constructor but base class does not have a default constructor?

- A) It compiles normally
- B) Compiler auto-invokes a private constructor
- C) **Derived class must explicitly call base class constructor**
- D) Derived object cannot be created

Q. Which query finds departments that have no employees?

- A) SELECT dept FROM Dept WHERE dept IN (SELECT dept FROM Emp);
- B) SELECT dept FROM Dept WHERE dept NOT IN (SELECT dept FROM Emp);
- C) SELECT dept FROM Dept d WHERE NOT EXISTS (SELECT * FROM Emp e WHERE d.dept = e.dept);
- D) SELECT dept FROM Emp GROUP BY dept;

Q. A graph contains negative edges but no negative cycle. Which algorithm works?

- A) Only Dijkstra
- B) Only Floyd-Warshall
- C) **Bellman-Ford and Floyd-Warshall**
- D) None

Q. In optimistic concurrency control, validation phase checks:

- A) Timestamp ordering
- B) Lock compatibility
- C) **Conflicts before commit**
- D) Shadow paging



Q. Which of the following is TRUE about object destruction?

- A) Destructor is overloaded
- B) Destructor can take parameters
- C) **Destructor is automatically called**
- D) Destructor cannot be defined

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Q. Time complexity of binary search recursive implementation:

- A) $O(\log n)$
- B) $O(n)$
- C) $O(n \log n)$
- D) $O(n^2)$



Q. Which constructor is used when an object is created from another object?

- A) Default constructor
- B) Static constructor
- C) **Copy constructor**
- D) Friend constructor

Q. Which method reads only the first 10 rows of a CSV?

- A) `pd.read_csv(file, head=10)`
- B) `pd.read_csv(file, nrows=10)`
- C) `pd.read(file, top=10)`
- D) `pd.csv_to_df()`

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Q. `df.loc['A']` selects rows based on:

- A) Integer index
- B) Boolean mask
- C) **Label-based index**
- D) Slice index



Q. What is the result of:

```
d = {'x':1, 'y':2}  
d.get('z', 5)
```

- A) Error
- B) 0
- C) 5
- D) 2

**Q. An attacker modifies data while it is being transmitted over the network.
Which attack is this?**

- A) MITM
- B) SQL Injection
- C) Ransomware
- D) Phishing

Q. What is with `open("file.txt")` as `f`: used for?

- A) Manual closing of files
- B) Automatic file handling & closing
- C) Opening only binary files
- D) Opening multiple files at once

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Q. In C, which function declaration is correct?

- A) `func(int a, int b);`
- B) `int func(a,b);`
- C) `int func(int,int);`
- D) `def func(a,b)`



Q. Which function adds an element at a specific index in python?

- A) **insert()**
- B) append()
- C) extend()
- D) add()



Q. Which field in the IPv4 header prevents packets from looping indefinitely?

- A) Version
- B) **TTL**
- C) Header checksum
- D) Identification



Q. Which operation reshapes a DataFrame by transforming row indexes into columns(long → wide format)?

- A) stack()
- B) **unstack()**
- C) transpose()
- D) pivot()

Q. plt.plot() is used to draw:

- A) Bar chart
- B) Scatter plot
- C) Line chart
- D) Heatmap

Q. In Java, which block is always executed?

- A) try
- B) catch
- C) throw
- D) **finally**



Q. A proxy firewall operates at which layer?

- A) Layer 2
- B) Layer 4
- C) Layer 7
- D) Layer 1